

ICRA 2026: What 2,951 Papers Say About Where Robotics Is Going

This analysis covers the 2,951 papers in the ICRA 2026 (Vienna, June 1–5) technical program — everything presented at the conference. That includes contributed papers accepted through ICRA's peer review (5,088 submissions; the official acceptance count is not yet published), journal papers (RA-L and similar) that opted into a conference presentation, 131 lightly-reviewed late-breaking results, and 36 award-session entries. "Papers" below means program entries, not ICRA acceptances. 311 of them (10.5%) carry a company affiliation. The keyword counts below are exact over the full corpus; the trend claims are synthesized from 30 independent full-abstract readings covering all papers. Where the readings disagree, the disagreement is flagged rather than averaged away.

Macro trends

1. Generative action policies won — and the field has already moved on to fixing them

Diffusion and flow-matching policies are now the unmarked default action layer for manipulation; the research frontier is no longer "does it work" but latency, guarantees, and mode collapse.

Evidence. Imitation Learning (196 of 2,951) and Learning from Demonstration (131) are the 4th and 9th most common keywords; "Imitation Learning + Learning from Demonstration" is the 2nd most frequent keyword pair (56 papers), and "Deep Learning in Grasping and Manipulation + Imitation Learning" the 3rd (47). All 30 batch readings independently report diffusion policies as the assumed baseline being patched, not the contribution. Exemplars of the repair work: *Mini Diffuser: Fast Multi-Task Diffusion Policy Training Using Two-Level Mini-Batches* (ThI1I.379) hits 95% of SOTA performance at 5% of training time; *Implicit Maximum Likelihood Estimation for Real-Time Generative Model Predictive Control* (TuI2I.301) matches diffusion planners at ~100x faster inference; *Shifted Flow Policy: Uncertainty-Aware Time Reparameterization for Visuomotor Learning* (WeBT2.4) attacks action chunking itself. Flow matching is visibly displacing diffusion in at least 8 of 30 batches, with the pio VLA now being RL-finetuned through its flow head (*Reinforcement Fine-Tuning of Flow-Matching Policies for Vision-Language-Action Models*, WeI1I.160). A counterintuitive landmark: *Dense-Jump Flow Matching with Non-Uniform Time Scheduling for Robotic Policies* (ThBT2.2) shows that adding inference integration steps universally *degrades* flow-matching policies.

What changed: Two years ago the debate was diffusion policy vs. ACT vs. plain behavior cloning. That debate is over. The new consensus treats the generative policy like the EKF circa 2010 — infrastructure you wrap with safety filters, uncertainty gates, and faster samplers — and the insurgent (flow matching) is already displacing the incumbent (DDPM-style diffusion) on speed.

2. Foundation models were demoted from robot brain to robot component — wrapped in verification

LLMs/VLMs/VLAs are everywhere, but the converged architecture keeps them out of the control loop: they do semantics, while classical planners, solvers, and verifiers keep the guarantees.

Evidence. AI-Based Methods (78) plus AI-Enabled Robotics (70) keywords understate it — every one of the 30 batch readings describes foundation models slotted into classical pipelines with explicit firewalls: LLM output compiled into PDDL/behavior trees/FSMs, VLM calls gated behind confidence triggers, plans checked by scene-graph verifiers (*VeriGraph: Scene Graphs for Execution Verifiable Robot Planning*, ThI2I.226; *PSALM-V: Automating Symbolic Planning in Interactive Visual Environments with Large Language Models*, ThI1I.250, which doubles ALFRED success over raw Claude-3.7 planning). For VLAs specifically the work is deployability surgery: *The Better You Learn, the Smarter You Prune* (ThI2I.69) cuts 59% of FLOPs from a VLA and improves success +2.9%; runtime MCTS, uncertainty introspection, and token-streaming runtimes wrap frozen VLAs (batch 29). Multiple batches explicitly cite LLM hallucination in geometric reasoning as the design driver for delegating paths to classical graph search (batch 20).

What changed: The 2023–24 consensus was that end-to-end VLAs would absorb the stack the way LLMs absorbed NLP. ICRA 2026 shows the opposite settlement: foundation models as one verified module among many, with the LLM-as-monolithic-planner pattern actively retreating (batch 19 describes LLMs "demoted from planner to one-shot semantic parser"). The LLM playbook that did transfer is post-training: preference alignment, RL fine-tuning, token pruning, and activation steering applied to robot policies.

3. The data supply chain is the new battleground — and scale alone is losing to curation and human video

The binding constraint moved from policy architecture to where demonstrations come from; the answers are human video, crowdsourcing, synthetic data engines, and aggressive curation — while the "just scale data" thesis took direct empirical hits.

Evidence. With 196 Imitation Learning + 131 LfD papers, data engineering is the modal contribution: Microsoft Research Asia converts 1M episodes of in-the-wild human hand video into VLA pretraining data with zero annotation (*Scalable Vision-Language-Action Model Pretraining for Robotic Dexterous Manipulation with Real-Life Human Activity Videos*, TuI2I.295); Meta learns multi-fingered policies from Aria smart-glasses recordings with no robot data at all (*Dexterity from Smart Lenses*, TuI2I.111); NVIDIA crowdsources 7,500+ demos across 9 countries in 5 days via smartphones (*COBALT*, ThI1I.238). Against naive scale: Li Auto's 30,000-hour study finds the imitation power law breaks in closed-loop evaluation (*Data Scaling Laws for Imitation Learning-Based End-To-End Autonomous Driving*, ThI2I.259); *The Curse of Precision* (ThI2I.162) derives a super-exponential demo-count law for precision manipulation; *Quality Over Quantity* (ThI2I.153) shows influence-function curation beats more data; and

Retargeting Matters (TuI2I.269) shows humanoid policy failures trace to retargeting artifacts, not rewards.

What changed: The 2024 default was "collect more teleop and scale the model." ICRA 2026's default is "never pay for teleop if you can mine human video, synthesize in sim, or curate what you have" — and at least five independent batches report rigorous negative results against brute-force scaling. Data quality, not data volume, is being treated as the moat.

4. 3D Gaussian Splatting completed its takeover as the default scene representation

In one conference cycle, 3DGS displaced NeRF (and is encroaching on TSDF/point-cloud maps) across SLAM, simulation, localization, and even planning — NeRF now appears mostly as the baseline being beaten.

Evidence. The perception/mapping cluster is the single largest block of the conference: Deep Learning for Visual Perception (253), Localization (163), SLAM (132), Mapping (115), with Localization+SLAM the most frequent keyword pair (59). At least 12 of 30 batch readings independently report 3DGS as the new default. The representation is now load-bearing infrastructure: *VINGS-Mono: Visual-Inertial Gaussian Splatting Monocular SLAM in Large Scenes* (ThI2I.30) runs kilometer-scale with 50M Gaussians; *MCGS-SLAM* (WeI2I.200) does the first multi-camera RGB-only 3DGS SLAM; Waabi's *SaLF: Sparse Local Fields for Multi-Sensor Rendering in Real-Time* (TuI2I.170) renders camera + LiDAR at 600+ FPS for AV simulation; *Real-Is-Sim* (ThI2I.56) runs policies inside a 60 Hz Gaussian digital twin that the real robot mirrors. GS is being differentiated, queried for signed distance (feeding CBFs), and fused with sonar, radar, events, and thermal.

What changed: NeRF was the assumed future of robot scene representation as recently as 2024. It lost on speed and editability before it ever became infrastructure. The new questions are engineering ones — pruning for embedded deployment, dynamic scenes, physics coupling — which is what a settled default looks like.

5. The classical/learning war ended in a merger: learning inside structure, structure around learning

RL-in-sim is commodity and diffusion is commodity, but pure end-to-end learning is being displaced by hybrids — residual RL over model-based controllers, RL tracking trajectory-optimization references, CBF/conformal safety wrappers — while classical optimization is resurging via GPU and solver engineering.

Evidence. Reinforcement Learning is the #2 keyword (250 of 2,951), but Optimization and Optimal Control (127) and Motion and Path Planning (229) are nowhere near dead, and "Legged Robots + Reinforcement Learning" (38 pairs) coexists with a model-based counterattack. The hybrid pattern is unanimous across batches: NVIDIA's *SPARR: Simulation-Based Policies with Asymmetric Real-World Residuals for Assembly* (ThI1I.266) adds real-world residuals to sim policies (+38.4%); *RAMBO: RL-Augmented Model-Based Whole-Body Control for Loco-Manipulation* (ThI2I.19) pairs QP feedforward

with RL feedback; *CBF-RL* (ThI1I.217) bakes safety certificates into training so a Unitree G1 needs no runtime filter. The classical resurgence is concrete: *Primal-Dual iLQR for GPU-Accelerated Learning and Control in Legged Robots* (WeI2I.387) beats acados by up to 700% and runs whole-body MPC for 16 robots in <25 ms; *Whole-Body Model-Predictive Control of Legged Robots with MuJoCo* (ThI1I.172) gets a full-size humanoid walking with plain iLQR and finite-difference derivatives, no learning; *Push Anything* (WeAT1.5) does open-world pushing at 98% success with contact-implicit MPC in an award session.

What changed: The 2022–24 narrative was learning replacing model-based control. The 2026 reality is division of labor: learned components generate and adapt; classical optimization executes, certifies, and increasingly wins outright on speed via GPU implementations. Conformal prediction and CBF-QPs have quietly become the standard safety vocabulary around any learned policy.

6. Touch and force became first-class citizens — vision-only manipulation was declared insufficient

Tactile and force sensing moved from sensor demos to the core of policy learning, with purpose-built data-collection hardware and explicit claims that vision-only demonstrations hide the contact events that matter.

Evidence. Force and Tactile Sensing counts 86 of 2,951 papers, plus Dexterous Manipulation (74); roughly a third of batch readings flag a dedicated tactile/force theme. The integration is structural, not bolt-on: *Spatially-Anchored Tactile Awareness for Robust Dexterous Manipulation* (TuAT3.1) anchors tactile features in the hand's kinematic frame for sub-millimeter bimanual USB-C mating; *Tactile-Conditioned Diffusion Policy for Force-Aware Robotic Manipulation* (TuI1I.229) makes grip force an explicit policy output; *ManipForce* (ThI2I.257) captures high-frequency F/T during human demos for 83% success on gear assembly. The striking datapoint: *FD-VLA: Force-Distilled Vision-Language-Action Model* (ThI2I.154) reports a force token hallucinated from vision outperforming the physical force sensor it imitates. Hardware capture rigs (UMI-style grippers with F/T sensors, exoskeleton gloves under \$700) are proliferating, and tactile simulation now runs 4,096 parallel RL environments on one consumer GPU (batch 17).

What changed: Tactile sensing was a perennial "promising modality" for a decade. What changed is the demand side: contact-rich manipulation policies hit a wall on vision alone, and the imitation-learning data pipeline created a concrete place to plug force signals in. Touch is now infrastructure on the same trajectory diffusion policies followed two years earlier.

The tensions

Learning vs. model-based — the field's loudest argument, with classical methods landing real punches. Against the learning tide: *LPV-MPC for Lateral Control in Full-Scale Autonomous*

Racing (TuBT2.5) holds a racecar stable above 160 mph with zero learning; *Reference-Free Sampling-Based Model Predictive Control* (ThI1I.109) gets emergent quadruped trotting, galloping, and backflips from CPU-only MPPI with no training; Motional's *TreeIRL* (ThI2I.216) puts MCTS planning on 500+ public-road miles and beats learned planners; *The Price Is Not Right* (TuI2I.336) shows PDDL + learned control beating fine-tuned pio 95% vs. 34% at ~100x less training energy; *Adios Stabilizers* (WeBT3.9) walks five humanoids with classical LIP MPC and no learned policy. The learning camp answers with results classical methods can't touch — zero-shot dexterous grasping from RGB, chicken deboning, champion-level vision racing. The synthesis (hybrids) is winning, but the boundary is genuinely contested, batch by batch.

Sim vs. real data — direct contradiction in the corpus. *Synthetic vs. Real Training Data for Visual Navigation* (WeI1I.74) finds sim-trained policies beating real-data-trained ones by 31 points, and zero-shot sim2real claims abound (98.6% quadrotor gate success from an editable GS sim, batch 24). Yet other papers reject simulation outright for un-simulatable domains (snow navigation learned directly in the real world, batch 22), Li Auto's scaling-law study (ThI2I.259) shows open-loop sim metrics don't predict closed-loop behavior, and the residual-RL school (SPARR, ThI1I.266) implicitly concedes sim alone isn't enough. The emerging compromise — train control in sim, adapt perception in real, or sim base + real residual — is an admission that neither side won.

VLA optimism vs. documented fragility. The same conference that treats VLAs as infrastructure also published the sharpest attacks on them: a single adversarially textured sphere collapses Pio/RDT success by 31–40% (*Exploiting Vulnerabilities*, WeI1I.148); visual clutter cuts VLA success up to 34% with low inter-model agreement (*Distracted Robot*, ThI2I.54); observation-space shift from preceding skills drops OpenVLA 54% with no standard remedy working (*BOSS*, ThI1I.17); NVIDIA's GRooT N1.6 falls from 90% to 4.5% on multi-step tasks (TuI1LB.21); a standalone VLA scored 9% on decomposable LIBERO-10 tasks until an LLM planner was bolted on top (WeI1LB.7). The field is simultaneously standardizing on VLAs and publishing evidence they memorize rather than understand. This tension is unresolved and is the most consequential one for anyone betting a company on generalist policies.

End-to-end vs. modular. End-to-end driving is in open retreat: oracle-perfect predictions don't improve planning and a rule-based lane-follower beats imitation-learned planners (*Perfect Prediction or Plenty of Proposals?*, TuI2I.72); learned planners' scores collapse under realistic reactive traffic agents while rule-based ones degrade gracefully (ThI1I.123); many end-to-end stacks show near-zero correlation between their plans and the navigation input they were given (WeI1I.284); explicit intention prediction beats the learned monolith for parking (TuI1I.193). In manipulation the same fight appears as dual-system (fast executor + slow reasoner) architectures vs. monolithic VLAs — with dual systems currently winning on results.

Diffusion vs. its own latency. Even the winning method class is at war with itself: more denoising steps can hurt (ThBT2.2), IMLE matches diffusion at ~100x speed (TuI2I.301), consistency models and one-step flows are displacing iterative sampling, and MIT showed diffusion policies only coarsely approximate the constraint manifolds in their training data while success metrics hide it (batch 12, WeI2I.233).

The white space

Security is almost absent — at the exact moment robots are being deployed. Across 2,951 papers, the security cluster is countable on two hands: an Internet-wide scan showing deployed ROS 2 robots are fingerprintable and exploitable in default configurations (*The Case of Metadata Leakage in ROS 2*, ThI1I.83), the first viewpoint-consistent 3D adversarial object against visuomotor policies (WeI2I.289), provably undetectable false-data-injection attacks on trajectory tracking (TuI1I.350), and a handful more. Given that the same conference is full of fleet deployments and humanoid pilots, the attack-surface-to-research ratio is the most lopsided in the corpus.

Evaluation rigor and reproducibility are a rounding error. *Unveiling Non-Reproducibility in LiDAR-Inertial Odometry* (ThI2I.417) shows SOTA results aren't reproducible run-to-run; a 5,040-trial cross-lab grasping benchmark (TuI2I.29) shows algorithm rankings flip with lighting and gripper choice; *Event-LAB* (ThI1I.84) shows event-camera comparisons weren't apples-to-apples. Statistically rigorous policy evaluation (Waymo's *Reliable and Scalable Robot Policy Evaluation with Imperfect Simulators*, ThI2I.206) exists but is rare. For a field claiming deployment readiness, almost nobody measures whether its own numbers replicate.

Energy and cost accounting barely exist. Two papers stand out precisely because they are alone: compute energy now rivals mechanical energy on a lightweight cobot (TuI2I.272), and the PDDL-vs-VLA comparison that metered training energy (TuI2I.336, ~100x difference). Amid foundation-model enthusiasm, almost no one reports joules, dollars, or BOM impact — strange for a field whose products must ultimately pencil out.

Reliability past the demo is conspicuously thin. NVIDIA's *Refinery* (ThI1I.129) explicitly frames ~80% research-grade success as inadequate for industry, and TRI's grocery-store field test (*SHOPPER*, WeI1I.104) reads as an honest catalog of unsolved grasping failures — but papers targeting the 99%+ regime, recovery behaviors, and long-duration autonomy are a tiny minority. A bibliometric paper at the conference itself (*Is AI Really Pervasive?*, TuI2I.91) concludes AI has not actually penetrated several core autonomy functions, deflating the everything-is-learned narrative from inside.

Humanoid hype vs. humanoid paper count. Legged Robots is keyword #19 (104 of 2,951, 3.5%), and "Humanoid and Bipedal Locomotion + Reinforcement Learning" appears in just 28 papers. The capital concentration in humanoids is wildly out of proportion to the share of the field's research output addressing them — most of the conference is still arms, cars, drones, and SLAM.

Industry watch

311 of 2,951 papers (10.5%) have a company affiliation. The portfolio tells you the roadmaps:

- **NVIDIA (24 papers, #1)** is building the sim-data flywheel end to end: data curation (*SCIZOR*, ThBT2.3), 53M-grasp generation (*GraspGen*, ThI1I.73), sim-to-real assembly at industry-grade reliability (*SPARR*, ThI1I.266; *Refinery*, ThI1I.129), VLA post-training (*GRAPE*, ThI1I.94) and runtime verification (ThI2I.128). The bet: own the synthetic-data and training infrastructure layer for everyone else's robots.
- **Toyota / TRI (14+ papers)** is the most methodologically diverse: video-diffusion data synthesis (*AnchorDream*, TuI2I.274), millisecond classical trajectory optimization with emergent drifting (WeI1I.303), real Supra drifting with honest negative results on online adaptation (ThI1I.349), humanoid teleop datasets, and brutally honest field tests (*SHOPPER*, WeI1I.104). Reads like a lab hedging across the entire learning-vs-model-based spectrum.
- **Huawei (~25–30 papers across entities)** is quietly assembling a full embodied-AI stack: a ~\$14k open bimanual humanoid (*OpenPyRo-A1*, WeI2I.356), demonstration-data surgery and synthesis (NICE, DRAW2ACT), VLA stress-testing benchmarks (*Distracted Robot*, ThI2I.54), VR humanoid teleop, and GS driving simulation. The broadest robotics surface of any company at the conference.
- **Waymo (4)** has pivoted research to the LLM era on its own terms: RL fine-tuning of multimodal LLMs for planning (*MAGNIFIED*, ThI2I.79), VLMs as preference raters and judges (DriveCritic), and statistically rigorous evaluation (*SureSim*, ThI2I.206) — governance and post-training, not new perception stacks.
- **Meta and Microsoft** are positioning consumer hardware as the robot-data acquisition layer: Aria smart glasses feeding manipulation policies with zero robot data (TuI2I.111), and MSRA's 1M-episode human-video VLA pipeline (TuI2I.295).
- **The Chinese humanoid cluster is the most coordinated industrial story at the conference.** Galbot (4 papers: motion tracking *Track Any Motions under Any Disturbances* ThI1I.179, GS navigation sim, sim2real grasping), Midea (HumanoidExo wearable data collection TuI2I.202, multisensory language-action models), ByteDance (ByteWrist hardware, shared-autonomy dexterous data engines, DemoBot ThI1I.107), Alibaba (RynnVLA-001 TuI1I.270), X-Humanoid, Fourier (open-sourced hand), Beijing Innovation Center (force-aware VLA fine-tuning, *CRAFT* ThI2I.249), Booster, and Unitree-as-substrate. Pattern: hardware + data-collection rigs + VLA post-training, published openly — they are commoditizing the humanoid stack layer by layer.
- **Automakers (Li Auto, Xiaomi EV, Hyundai, Mercedes, Honda — ~20 papers combined)** converge on one theme: making foundation models cheap and trustworthy enough for production — token pruning, 0.9B VLMs, uncertainty-gated fast/slow stacks (*FASIONAD*, TuI1I.295), camera-only distillation, and scaling-law due diligence before spending more on fleet data (ThI2I.259).
- **Amazon (10+4)** stays on warehouse economics (bin packing, contact-implicit MPC for non-prehensile manipulation, probabilistic safety certificates); **MERL (9)** sells advanced control and tactile estimation as IP; **Boston Dynamics AI Institute / RAI (14)** is pushing sampling-based

MPC tooling (judo) and hybrid QP+RL whole-body control (*RAMBO*) — notably model-based-leaning for a learning-era lab.

Humanoid deep-dive: the 2026 recipe for making a humanoid walk

The recipe converged hard this year. Across 28 "Humanoid and Bipedal Locomotion + RL" papers and the humanoid signals in all 30 batch readings, the consensus stack is:

1. Buy the platform; the Unitree G1 is the new Franka. The G1 appears as the default validation hardware in well over a dozen papers across at least 10 batches — including non-locomotion work that just needs a humanoid demo (WeI1I.122) — with the Booster T1 (~23 DoF, used for table tennis *PACE* TuI2I.206 and soccer kicking *MAKP* TuI2I.158) as the budget challenger, and even MATLAB/Simulink toolchains shipping for the G1 (ThI2LB.4). Hardware differentiation has moved to subsystems: parallel-linkage knees/ankles that need dedicated models (*Control of Humanoid Robots with Parallel Mechanisms Using Differential Actuation Models*, TuI1I.90; MPPI over learned ankle joint maps, TuI1I.258), cycloidal/quasi-direct-drive actuators, and dexterous hands whose cost floor fell below \$200 in structure cost.

2. Get the actuator model right — it, not the simulator, is the sim2real bottleneck. A 300+ kg hydraulic quadruped walked because of a sub-microsecond *analytical* actuator model that beat neural actuator nets (ThI2I.378); friction-aware actuator identification (WeI2LB.3) and uncertainty-guided actuator-data collection (batch 29) make the same point. Domain randomization is the fallback, not the fix.

3. Retarget human motion, then train massively parallel RL tracking in sim — and obsess over data quality, not rewards. The canonical pipeline is human MoCap/video → retargeting → PPO in Isaac-style sim with teacher-student distillation → zero-shot deployment (*ExBody2*, TuI1I.76; *Track Any Motions under Any Disturbances*, ThI1I.179). The 2026 insight is that retargeting quality is the moat: *OmniRetarget* (WeAT1.4) gets 30-second parkour on a G1 with only 5 reward terms and no curriculum by fixing the data, and *Retargeting Matters* (TuI2I.269) shows open-source retargeting nearly matches Unitree's closed dataset. Reward engineering is being replaced by structure: CLF-shaped rewards from classical gait references (*CLF-RL*, TuBT3.8), and even LLM-authored training curricula deployed zero-shot (AURA, TuI2I.284).

4. One tracking policy, many skills — with model-based control re-entering through the side door. Single whole-body controllers now span minute-long dynamic sequences (*Kungfubot2*, ThI1I.251; *Towards Adaptable Humanoid Control Via Adaptive Motion Tracking*, WeBT3.5), and skill graphs schedule transitions at runtime. But the pure-RL monopoly is over: *Adios Stabilizers* (WeBT3.9) walks five different humanoids with closed-loop LIP MPC and no learning; plain iLQR-MPC in MuJoCo controls a full-size humanoid in real time (ThI1I.172); full-order inverse-dynamics MPC runs at 80 Hz on open tooling (WeI2I.426); and the hybrid — RL parametrizing an MPC (ThI1I.241) or correcting a QP feedforward (*RAMBO*, ThI2I.19) — is the fastest-growing recipe.

5. Engineer for falling, not just against it. Falls became a first-class skill: unified fall-prevention/impact/get-up policies (WeI1I.114), Disney's stylized controlled falls (TuI1I.218), real-time fall prediction with 1.1 s lead time on Digit (TuI2I.113), and fault-tolerant locomotion through joint locking and power loss (*TOLEBI*, WeBT3.4). Safety certificates moved inside training (*CBF-RL*, ThI1I.217: G1 stairs with no runtime filter).

6. The data flywheel is the business model. Whole-body demonstration collection is commoditizing fast: TWIST2's mocap-free VR rig collects 100 demos in 15 minutes (WeBT3.1), Midea's wearable exoskeleton teaches walking from human-worn data alone (TuI2I.202), and 260+ hours of humanoid motion auto-extracted from human video shows clean scaling laws (ThI2I.155). Cross-embodiment locomotion checkpoints (*H-Zero*, WeI1I.168; *EAGLE*, ThI1I.110 — one policy across Unitree H1/G1 and Fourier N1) mean a new humanoid design can inherit walking nearly for free.

The headline demos prove the recipe's ceiling rose: *HITTER* (WeBT1.7) sustains 106 table-tennis rallies against a human with sub-second reaction. The honest caveat from the same corpus: manipulation reliability, VLA fragility, and the absence of cost/energy/security work mean the walking problem is converging just as the *useful work* problem takes its place.

Generated 11 June 2026 from the official ICRA 2026 program (ras.papercept.net) — 2,951 program entries, 30 independent full-abstract readings + exact corpus statistics.